

BEST COMPROMISE BETWEEN TELE SUPPLY AND MODULATED RATE OF BACK SCATTERED SIGNAL IN RFID SYSTEM

Alexandre Tramoni, Claude Tetelin

Alexandre Malherbe, Jerome Conraux

University of Provence, L2MP-CNRS-ISEN
Computing/Electronics
Place Pompidou, 83000 Toulon, France

ST MICROELECTRONICS
Smart Card Division
13119 Rousset, France

ABSTRACT

Emerging applications of contact less smart card operating at 13.56 MHz like video transmission and ICAO e-passport need high data rates. This study focuses on the communication from the deported element (batteryless smartcard or transponder) to the reader. In order to limit the cost of the transponders, most of the time, smart cards are batteryless. The chip is connected to an antenna and is supplied by the power coming from the induced voltage of the coupled antennas [1]. Designers have to deal with a compromise between best chip supply and best load modulation. The purpose of this paper is to present a new method to compute the best load to switch for a given power supply. For this supply, the best modulation degree of the back scattered signal will be reached. The theoretical results and the simulation results will be compared to measurements.

1. INTRODUCTION

Contactless smart cards systems are characterized by three elements: the reader, the card (or transponder) and the transmission channel. The reader and card antennas are, generally, loops antennas with different geometries depending on the application. By measuring S-Parameters of the coupled antennas it has been demonstrated [2] that we can infer the exact value of the coupling coefficient, k , between antennas. This coefficient is used to have an electric equivalent circuit of the complete system [3]. The relation between k and the mutual inductance value, M , is given by (1).

$$M = k\sqrt{L_L L_C} \quad (1)$$

L_L and L_C are respectively the inductances values of the reader antenna and the card antenna. Fig.1 shows the model of the contact less system used for simulation.

Generally, reader antennas are matched to 50 ohm at 13.56 MHz. The matching circuit is made of two capacitances C_{R1} and C_{R2} [4]. R_{LR} is the serial resistance of the loop reader antenna and R_R is a resistance added to the circuit in order to achieve a convenient quality factor for the

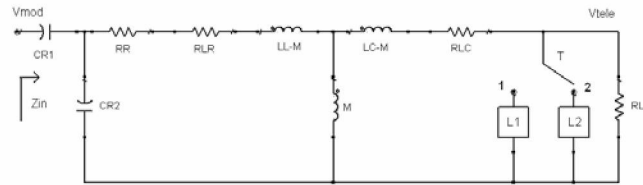


Fig. 1. Contactless system

reader antenna. R_{LC} is the serial resistance of the loop card antenna and R_L simulates the input impedance of the chip connected to the card antenna. In order to send data to the reader, an electronic switch is controlled by the chip. L_1 and L_2 are two complex loads. Whether L_1 or L_2 is connected, the quality factor of the card antenna is changed. The consequence is that the input impedance of the system " Z_{in} " is changed and back scattered signal can be demodulated. In this paper L_1 and L_2 will be determined to achieve the best modulation degree of back scattered signal " V_{MOD} " with a given minimum telesupply voltage " V_{TELE} ".

2. INPUT IMPEDANCE, V_{TELE} AND V_{MOD}

In this paper, all the demonstrations and calculations will be made with the equivalent impedance " Z_{load} " of L_1 and L_2 depending on the state (T_1 or T_2) of the switch T on Fig.1. As these loads can be reactive or/and resistive, Z_{load} will be given by the following expression.

$$Z_{load} = (A_i + jB_i)\Omega \quad (2)$$

Where A_i is the resistive part of L_i and B_i is the reactive part of L_i when the switch T is in the state T_i . First of all, input impedance Z_{in} has to be expressed as a function of Z_{load} .

$$Z_{in} = \frac{1}{jC_{R2}\omega + \frac{1}{\alpha + \beta}} \quad (3)$$

With:

$$\alpha = \frac{jM\omega(\frac{R_L Z_{load}}{R_L + Z_{load}} + R_{LC} + j(L_C - M)\omega)}{jM\omega + \frac{R_L Z_{load}}{R_L + Z_{load}} + R_{LC} + j(L_C - M)\omega}$$

$$\beta = R_R + R_{LR} + j(L_L - M)\omega$$

Then, back scattered voltage " V_{MOD} " is simply given by (4).

$$V_{MOD} = \frac{Z_{in}}{Z_{in} + R} V_{in} \quad (4)$$

R is the serial internal resistance of the generator delivering the voltage V_{in} at the frequency of 13.56 MHz. To follow, the voltage used to supply the chip, V_{TELE} , is expressed by (5).

$$V_{tele} = \varphi \left[\gamma - jC_{R1}\omega\varepsilon + j\frac{\varepsilon - \beta\gamma - jC_{R2}\omega\varepsilon}{M\omega} \right] \quad (5)$$

With:

$$\varphi = \frac{R_L Z_{load}}{R_L + Z_{load}}; \gamma = \frac{V_{in} - V_{MOD}}{R}$$

and

$$\varepsilon = V_{MOD} + j\frac{\gamma}{C_{R1}\omega}$$

As the voltages V_{MOD} and V_{TELE} are defined by the electrical characteristics of the system and the switched load, it is possible to choose Z_{load} in order to achieve a modulation degree for a given supply voltage of the chip. Therefore in this article the minimum voltage to supply the chip will be denoted $V_{supply-min}$.

3. CHOICE OF THE SWITCHED LOAD

On the Fig.2, the magnitudes of V_{MOD} and V_{TELE} are plotted on the same graph. A_i is the real part of Z_{load} and B_i is the imaginary part.

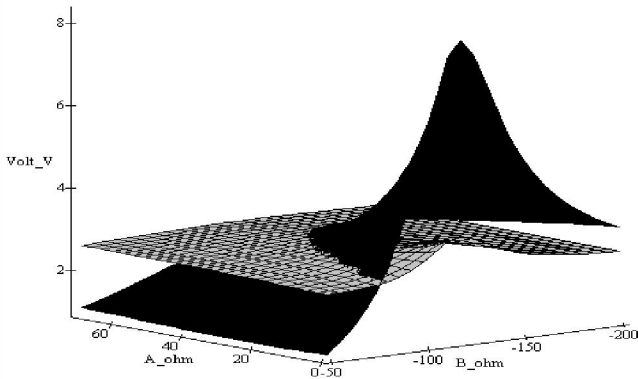


Fig. 2. V_{TELE} (black) and V_{MOD} (grey)

The modulation degree of back scattered signal (M) is define by (6).

$$M = \frac{\max(|V_{MOD}(A_i, B_i)|) - \min(|V_{MOD}(A_i, B_i)|)}{\max(|V_{MOD}(A_i, B_i)|) + \min(|V_{MOD}(A_i, B_i)|)} \quad (6)$$

In order to optimise the compromise between tele supply of the chip and the modulation degree of the back scattered signal, the following requirements have to be fulfilled. Considering $E \in \mathbb{R}^2$, a set of A_i, B_i such

$$V_{TELE}(A_i, B_i) \geq V_{SUPPLY-MIN}$$

Then,

$$(A_1, B_1) \in E \text{ and we have:}$$

$$V_{MOD}(A_1, B_1) = \max(|V_{MOD}(A_i, B_i)|)$$

$$(A_2, B_2) \in E \text{ and we have:}$$

$$V_{MOD}(A_2, B_2) = \min(|V_{MOD}(A_i, B_i)|)$$

The best compromise is achieved when these conditions are satisfied and the maximum modulation degree is given by (7).

$$M = \frac{V_{MOD}(A_1, B_1) - V_{MOD}(A_2, B_2)}{V_{MOD}(A_1, B_1) + V_{MOD}(A_2, B_2)} \quad (7)$$

4. DETERMINATION OF L_1 AND L_2

The aim of this part is to apply the previous method to a concrete case. $V_{supply-min}$ will be chosen to be equal to 3V with an error of 10 percents. L_1 and L_2 values will be found for this considering case.

$$L_1 = (A_1 + jB_1)\Omega \text{ and } L_2 = (A_2 + jB_2)\Omega \quad (8)$$

In this part, the contact less system is characterised by:

$$L_L = 1350nH; R_{LR} = 1ohm; L_C = 1379nH$$

$$R_{LC} = 0.74ohm; R_R = 15ohms; R_L = 1200ohms$$

On the Fig.2, magnitudes of V_{TELE} and V_{MOD} have been plotted with these parameters. By applying the method A_1 and A_2 are equal to 0. The maximum variation of V_{MOD} is obtained for two pure reactive loads. On Fig.3, V_{MOD} and V_{TELE} are plotted in only two dimensions for $A_i = 0$.

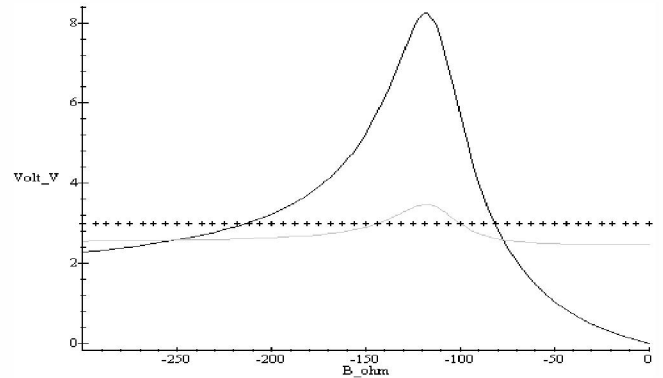


Fig. 3. V_{TELE} (black) and V_{MOD} (grey)

The dots line represents the desired minimum supply voltage $V_{supply-min}$. First of all, B_1 corresponds to the maximum value of V_{MOD} and to the maximum value of V_{TELE} and $B_1 = -117.49\Omega$. So, the load impedance L_1 is a capacitance (C_{L1}) and his value is 99pF at 13.56 MHz. At the operating frequency, C_{L1} and L_C make a resonant circuit as demonstrated by (9).

$$f = \frac{1}{2\pi\sqrt{L_C C_{L1}}} \quad (9)$$

Finding B_2 needs a little more attention. The method proposed in the previous section leads to two possible equivalent values for B_2 .

$$B'_2 = -213.62\Omega \text{ and } B''_2 = -81.68\Omega$$

For these two values we have $V_{TELE} = 3V$. In order to choose between these values it is necessary to have a look on V_{MOD} . Unfortunately, the condition on V_{MOD} is not sufficient because $V_{MOD} = 2.62V$ for both loads. The following section explains how the choice between B'_2 and B''_2 has to be done.

5. PROCESS VARIATIONS

B'_2 and B''_2 correspond respectively to capacitances (C'_2 and C''_2) whose values at the operating frequency of 13.56 MHz are 54.94pF and 143.69pF. As these two capacitances gives the same values for V_{MOD} and satisfy to the conditions of part.3, another criterion has to be taken into account. Depending on technologies, process variations could reach more than 10 percents of the initial capacitances values. Tab.1 shows the variations of V_{TELE} and V_{MOD} for 10 percents variations of C'_2 and C''_2 . The impedance B''_2 makes

	C'_2	C''_2
V_{TELE}	3V	3V
V_{MOD}	2.62V	2.62V
$\Delta V_{TELE} (C2 \pm 10\%)$	0.7V	1.79V
$\Delta V_{MOD} (C2 \pm 10\%)$	0.07V	0.19V
$\min(V_{TELE})$	2.74V	2.2V

Table 1. Process Variations

the contact less system more sensible to the process variations. The minimum telesupply voltage could be down to 2.2V which is less than $V_{supply-min}$. For this reason, B_2 will be chosen equal to $B'_2 = -213.62\Omega$. To conclude this section:

$$L_1 = -j11.49\Omega \text{ and } L_2 = -j213.62\Omega$$

The modulation degree of the back scattered signal is computed by (7) and is equal to:

$$M = 13.9\%$$

6. S-PARAMETERS

Fig.1 shows the mutual inductance model. Coefficient coupling "k" is obtained by measuring the S-Parameters of the coupled antennas [2]. This section explains that it is not necessary to use the "mutual inductance" model. We will show here how modulation degree of backscattered signal can be related to S-Parameters measurement of the coupled antennas. The input impedance can be expressed as a function of S11 (10) where Z_C is the characteristic impedance of 50ohms [5].

$$Z_{in} = Z_C \frac{S11 + 1}{1 - S11} \quad (10)$$

By using this formula, we can derive V_{MOD} using the value of S11 parameter (11)

$$V_{MOD} = \left| \frac{V_{in} Z_C (1 + S11)}{R(1 - S11) + Z_C(1 + S11)} \right| \quad (11)$$

As V_{MOD} depends on S11 parameter, this is the same for the modulation degree of the backscattered signal. With $V_{in} = 5V$, the expression of M is:

$$M = \frac{|S11(A_1, B_1) + 1| - |S11(A_2, B_2) + 1|}{|S11(A_1, B_1) + 1| + |S11(A_2, B_2) + 1|} \quad (12)$$

The greater the module of S11, the greater V_{MOD} . In order to obtain the maximum modulation degree, the load commutation has to induce the maximum variation of the module of S11. Basically, V_{TELE} is a function of S11 and had to be considered. As a result, the module of S11 can not be down to 0. Fig.4 shows the variation induced by the load modulation for the example of the fourths section.

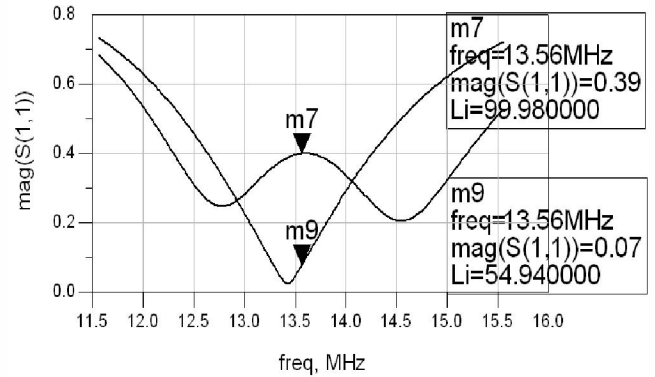


Fig. 4. Variation of magnitude of S11

7. SIMULATIONS, THEORY AND MEASUREMENTS

The aim of this part is to compare the theory with the simulation. The measurements and the simulation results will

be displayed on the same figure. Fig.5 shows the measurements and the simulation of the back scattered signal in the time domain. The switch rate is 424 kHz. The switch used for the measurements is not ideal and the measured R_{on} resistance is equal to 50ohms. Then, the simulation is performed with an electrically equivalent switch. The modula-

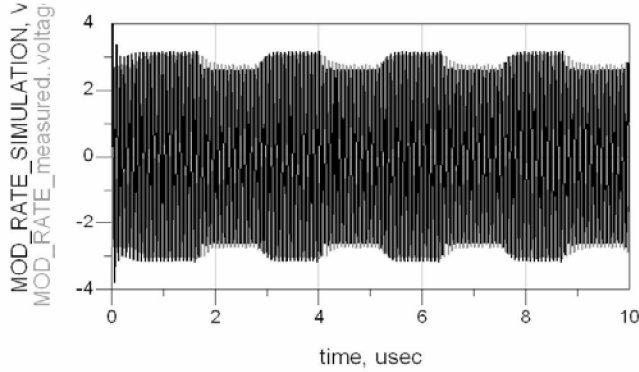


Fig. 5. Back scattered signal

tion degree is almost the same. The tele supply voltage is greater than 3V for simulation, theory and measurements. Tab.2 shows the modulation degree of the back scattered signal. Theory and simulation have been performed without

	Theory	Simulation	Measurement
Mod-Degree	9.22%	9.22%	7.08%

Table 2. Modulation degree

taking in account the quality factor of the switched capacitances. This can explain the differences between theory and simulation and measurement in Tab.2.

8. CONCLUSION

In this article, the best modulation degree of the back scattered signal is reached for a fixed minimum tele supply voltage. The complete contact less system is simulated and the equivalent circuit is given. The proposed method allows the RF designers to correctly predict what will really happened in measurements. The precision of this method allows to take in account the parasitic effects due to the implementation and process. By applying some variations to the components of the equivalent model, it is possible to choose the electrical appropriate properties of the antennas to fit the desired system characteristics. An interesting new study is to have a look on the "stagger tuning". This method consists on emitting another frequency than 13.56MHz [6].

9. REFERENCES

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